

EE 223P: Reverse Engineering

Course Code	EE 223P
Course Name	Reverse Engineering
L-T-P-C	1-0-0-1
Prerequisites	Faculty approval
Intended for	BTech EE/BTech VLSI
Mutual Exclusion	NA
Approval	53rd BoA (Dec 2023)

Course Contents

- **Topic 1:** Introduction to Reverse Engineering, need of Reverse Engineering, understanding of Reverse Engineering through examples, methodologies for Reverse Engineering, Reverse Engineering vs Forward Engineering, steps for Reverse Engineering (1 hour).
- **Topic 2:** Mechanical components used in joining (screw fasteners, bolts, nut and washers, compression fittings, pop rivets and rivets), and power transmission (belt drives, chain drives, gear drives and couplings), mechanical tools used in mechanical workshop: measuring tool, fastening-wrenches, fastening screwdrivers, fastening-pliers, cutting tools and striking/driving tools, standards of measurement: primary, secondary, tertiary and working standards, measurement devices: caliper and bevel protector (1 hour).
- **Topic 3:** Top down approach, product information through examples, important electrical/electronics devices: resistor, capacitor, inductor, diode, transistor, operational amplifiers and integrated circuit or microchip, various analysis and quantities, measuring instruments: multi-meter, function generator, oscilloscope, probe and connectors (1 hour).
- **Topic 4:** Introduction to engineering drawing, importance of engineering drawing, engineering drawing projections, detailed drawing, assembly drawing and its classification, exploded assembly drawing, advantages of exploded views (1 hour).
- **Topic 5:** Importance of engineering drawing in Reverse Engineering, engineering drawing of bench wise, engineering drawing standards in dimensioning, sectional view, tolerance, surface finish and welding, standard codes, engineering drawing format, production drawing of bench wise, Bill Of Material (BOM) (1 hour).
- **Topic 6:** Introduction to engineering materials, evolution of engineering materials, classification of engineering materials into metals, non-metals, ceramics and composites, features, identification, examples and applications of different materials, understanding through case studies, introduction to manufacturing processes, classifications of machining and joining process, introduction to casting process, various allowances and types of casting process (1 hour).
- **Topic 7:** Demonstration exercise: Reverse Engineering of Electric Kettle (1 hour).

Laboratory Modules

- 4 lab activities will follow the 7 hours indicated above. In the labs, the students will be disassembling and reassembling some day-to-day devices based on the lectured principles, and provide a report that will typically include
- Explanation of the working mechanism,
- Exploded drawing of the product,
- Assembly and disassembly procedure,
- Bill of Materials (BOM),
- Manufacturing processes and scope of materials,
- Discussion and Scope for Improvement.
- The devices used will be (but not limited to) torch, calculator, mouse and DC motor.

Text Books

1. K. Otto and K. Wood, **Product Design: Techniques in Reverse Engineering and New Product Development**, Prentice Hall, 2001.
2. Raja and Fernandes, **Reverse Engineering: An Industrial Perspective**, Springer- Verlag, 2008.

References

1. Eldad Eilam, **Reversing: Secrets of Reverse Engineering**, Wiley, 2005.
2. Chris Eagle, **The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler**.